Lec 33: Rotations and Angular Momentum

As we alluded to earlier, $\hat{J}$ is the generator of rotations.

Big difference is that 3D rotations do not commute. $R_x(\theta) R_y(\theta) \neq R_y(\theta) R_x(\theta)$.

S&N Sec. 3.1 – 3.3

See also Shankar ch. 12. and ch. 14 for spin.

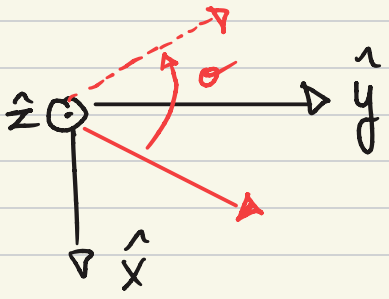
Recall in 3D space rotations are described by 3×3 real matrices:

$$R^T R = R R^T = \mathbb{1} \quad \text{w/} \quad R \cdot \vec{V} = \vec{V}'$$

e.g. $$R_z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

These are active rotations → rotate vector components and not the axes.

Counter clockwise rotation when seen from $+\hat{z}$:



Now for small $\theta = \varepsilon \ll 1$: $R_z(\varepsilon) \simeq \begin{pmatrix} 1-\frac{\varepsilon^2}{2} & -\varepsilon & 0 \\ \varepsilon & 1-\frac{\varepsilon^2}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} + O(\varepsilon^3)$

Now they approximately commute:

$$R_x(\epsilon) R_y(\epsilon) - R_y(\epsilon) R_x(\epsilon)$$

$$\simeq \begin{pmatrix} 0 & -\epsilon^2 & 0 \\ \epsilon^2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = R_z(\epsilon^2) - \mathbb{1} + O(\epsilon^3)$$

↳ higher order

- Rotations commute at $O(\epsilon)$
- commutator is itself a rotation at $O(\epsilon^2)$

In fact:

$$\left[R_x(\epsilon), R_y(\epsilon) \right] \cong -\left(\mathbb{1} - R_z(\epsilon^2) \right)$$

Note how $R(\epsilon)$ is the identity plus a small correction. Write it as:

Infinitesimal Rotation:

$$R_k(\epsilon) = \mathbb{1} - i\epsilon G_k$$

G_k → "generator" of rotations

Same logic as translation operator:

$$\int (dx) = \mathbb{1} - i \frac{dx}{\hbar} \hat{p}$$ → generator of translations

Note: $G^\dagger = G$ hermitian so that:

- $R_k(\epsilon) R^T(\epsilon) = \left(\mathbb{1} - i\epsilon G_k \right)\left(\mathbb{1} - i\epsilon G_k^T \right)$

$$\simeq \mathbb{1} - i\epsilon \underbrace{\left(G_k + G_k^T \right)}_{=0 \text{ if } G \text{ antisymmetric}} + O(\epsilon^2)$$

$$G_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\lambda = 0, \pm 1$.

But also G_k is Hermitian

Commutation relation:

$$[R_x(\epsilon), R_y(\epsilon)] = -\left(1 - R_z(\epsilon^2)\right)$$

$$\epsilon^2 [G_x, G_y] = i\epsilon^2 G_z \Rightarrow [G_x, G_y] = iG_z$$

you can keep doing this to show that:

1) $(G_k)_{ij} = -i\varepsilon_{ijk}$ with $\varepsilon_{012} = +1$ antisymmetric tensor.

2) $[G_i, G_j] = i\varepsilon_{ijk} G_k$

Looks familiar! Note $G_x^2 + G_y^2 + G_z^2 = 2 \times \mathbb{1}$

What is a finite rotation then?

$$R_k(\theta) = \lim_{N\to\infty} R_k(\theta/N)^N = \lim_{N\to\infty}\left(1 - iG_k\frac{\theta}{N}\right)^N$$

$$= e^{-i\theta G_k}$$

Alternatively, note that $G_x^2 = -\begin{pmatrix}1&&\\&1&\\&&0\end{pmatrix}$; $G_x^3 = G_x$

$$e^{-i\theta G_k} = \mathbb{1} - i\theta G_k - \frac{\theta^2 G_k^2}{2!} + \frac{\theta^3 G_k^3}{3!} + \frac{\theta^4 G_k^4}{4!}$$

$$= \begin{pmatrix}0&&\\&0&\\&&1\end{pmatrix} + \begin{pmatrix}1&&\\&1&\\&&0\end{pmatrix}\left(1 + \frac{\theta^2}{2} + \frac{\theta^4}{4!} + \dots\right)$$

$$+ \begin{pmatrix}0&-1&\\1&0&\\&&0\end{pmatrix}\left(\theta + \frac{\theta^3}{3!} + \dots\right)$$

$$= R_x(\theta)$$

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Another useful formula is

Rodrigues Rotation Formula:

$$R_{\hat{n}}(\theta)\,\vec{v} = \vec{v}\cos\theta + (\hat{n}\times\vec{v})\sin\theta + \hat{n}(\hat{n}\cdot\vec{v})(1-\cos\theta)$$

Quantum Analogs:

$$\hat{D}(\hat{n}, d\theta) = \mathbb{1} - \frac{i}{\hbar}(\hat{J}\cdot\hat{n})\,d\theta$$

"Drehung" → rotation

$$|\alpha\rangle_R = \hat{D}(R)\,|\alpha\rangle$$

Acts on Hilbert space.

Note $\hat{G}_k = \hat{J}_k/\hbar$ (recall $\hat{K} = \hat{p}/\hbar$)

Not a angular momentum in the sense

$$\hat{J} \neq \hat{x}\times\hat{p}$$

In general:

$$\hat{J} = \hat{L} + \hat{S}$$

$\hat{S}$ → Spin

$$D(\hat{n},\theta) = \lim_{N\to\infty}\left(\mathbb{1} - \frac{i}{\hbar}(\hat{J}\cdot\hat{n})\frac{\theta}{N}\right)^N = e^{-i\theta\frac{(\hat{J}\cdot\hat{n})}{\hbar}}$$

finite rotations.

Note we assume axes of rotations stays constant throughout. If not, then we'd need something similar to the time-ordered exponential.

Rotations form a group: $SO(3)$

- Identity element: $R \cdot \mathbb{1} = R \rightarrow \hat{D}(R) \cdot \mathbb{1} = \hat{D}(R)$
- Closure: $R_1 R_2 = R_3 \longrightarrow \hat{D}(R_1)\hat{D}(R_2) = \hat{D}(R_3)$
- Inverse: $R R^{-1} = R^{-1} R = \mathbb{1}$

  $\hookrightarrow \hat{D}(R)\hat{D}^{-1}(R) = \hat{D}^{-1}(R)\hat{D}(R) = \mathbb{1}$
- Associativity: $R_1(R_2 \cdot R_3) = (R_1 R_2) R_3$

  $\hookrightarrow \ldots$

BTW: $\hat{D}(R)$ is unitary: $\hat{D}^\dagger \hat{D} = \mathbb{1}$

$$\hat{D}(R_{\hat{n}}(\theta)) = \exp\left(-i \frac{(\hat{n}\cdot\vec{J})}{\hbar}\theta\right); \quad \vec{J} = (\hat{J}_x, \hat{J}_y, \hat{J}_z)$$

BTW 2: $SO(3)$ is a non-abelian group. This just means its elements do not commute.

Generators obey:

$$[\hat{J}_i, \hat{J}_j] = i\hbar\,\varepsilon_{ijk}\hat{J}_k$$

$\hat{J}_i$ span the Lie Algebra of $SO(3)$, i.e. $\mathfrak{so}(3)$.

Question: using $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}\hat{\mathbb{1}}$

show that $\hat{L} = \hat{\vec{r}} \times \hat{\vec{p}}$ satisfies

$$[L_x, L_y] = i\hbar \hat{L}_z .$$

Question 2: Show that

$$\left(\mathbb{1} - \frac{i}{\hbar}\hat{L}_z d\theta\right)|x, y, z\rangle = |x - y\,d\theta,\ y + x\,d\theta,\ z\rangle$$

using translation operator $\hat{T}(dx) = \mathbb{1} - \frac{i\hat{p}dx}{\hbar}$

Question 3: Show that $\hat{J}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2$ commutes with J_i for any $i = x, y, z$.

Question 4: Show that

$$e^{\frac{i}{\hbar}\hat{J}_z\theta}\,\hat{J}_x\,e^{-\frac{i}{\hbar}\hat{J}_z\theta} = \hat{J}_x\cos\theta - \hat{J}_y\sin\theta .$$

i.e: $\hat{D}^{+}(R)\,\hat{J}_i\,\hat{D}(R) = R_{ij}\hat{J}_j$

Vector Operator. because it transforms as a vector under rotations.